



# Nazmi Yücel Çan

**JUNIOR MACHINE LEARNING ENGINEER** • Türkiye (Open to Relocation & Remote)

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## TECHNICAL PROFILE

Junior Machine Learning Engineer specializing in model development, optimization, and edge AI deployment. Reduced mT5 SLM model size 5x (from 500MB to 100MB) via pruning, 8-bit LoRA, and INT8 quantization. Following computer vision experience, currently focusing on Natural Language Processing (NLP). Co-authored a research paper published in El-Cezeri Journal (2026). Developed Zettelkasten AI Notes (Pupulica AI Hackathon Top 10 Finalist), featuring a Two-Tier Vector RAG pool, CPU cross-encoder reranking, and Vulkan GPU local GGUF inference.

## SKILLS

### Programming Languages

Python (Advanced), SQL, Bash/Shell Scripting

### ML & AI Frameworks

TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face (LoRA/PEFT), ONNX Runtime, llama-cpp-python

### Core Domains

Computer Vision (OpenCV, YOLOv8-seg), NLP (Transformers, RAG & Embeddings), Data Preprocessing, Validation & Augmentation, Model Optimization (Quantization, Pruning, LoRA/PEFT), Edge AI & Inference Optimization

### Tools & Deployment

Git/GitHub, Linux (Ubuntu/Debian), REST API & Model Serving Fundamentals, PyQt5, Flet, Google Gemini API

**Languages:** Turkish (Native), English (Upper-Intermediate)

## EXPERIENCE

### Machine Learning Intern at Deep Think Technology

02/2026 – 06/2026

- Developed an end-to-end industrial "Apple Detection and Quality Analysis System", integrating YOLOv8-seg instance segmentation, illumination-robust CIEDE2000 color difference formula, and physical dimension measurement with the industrial camera SDK into a PyQt5 desktop application.
- Trained and deployed YOLOv8-seg models for real-time edge inference on resource-constrained embedded hardware, optimizing the image processing pipeline to eliminate data transfer bottlenecks and maintain continuous live stream processing.
- Engineered data preprocessing and augmentation workflows for industrial camera feeds, conducting systematic model validation to ensure operational stability under challenging lighting conditions.

## PROJECTS

### mT5 Bidirectional EN-TR Machine Translation (ONNX INT8 & ORT Mobile) [Kaggle]

(Python, PyTorch, Transformers, LoRA, Fine-Tuning, ONNX Runtime, NLP)

- Fine-tuned an mT5-small model with 8-bit LoRA for bidirectional English-Turkish translation, achieving a 0.6245 METEOR score. Reduced model size 5x (from 500MB to 100MB) and optimized performance for resource-constrained environments by applying model pruning, ONNX compilation, and INT8 quantization for mobile deployment via ONNX Runtime (ORT Mobile).

### Zettelkasten AI Notes (Pupulica AI Hackathon — Top 10 Finalist) [GitHub]

(Python, Flet, SQLite, RAG, llama.cpp, Gemini API, NLP)

- A privacy-first desktop knowledge management application implementing the Zettelkasten method using Python and Flet. Features a Dual AI Engine (Google Gemini and Vulkan-accelerated local GGUF models) with a Two-Tier In-Memory Vector RAG Pool, CPU cross-encoder reranking (Qwen3), and local embeddings (Harrier-0.6B) to synthesize atomic notes without duplicate concepts. Includes an interactive canvas mind map, live Markdown editor with LaTeX, smart WikiLinks, and SQLite WAL persistence.

### Arduino PC System Monitor [GitHub]

(Python, Arduino, TinyML, scikit-learn)

- A TinyML-powered system that monitors PC hardware resources (CPU, RAM, GPU, VRAM) and automatically optimizes power consumption. The system extracts hardware telemetry in real-time, executing an on-device SVM classification model on an Arduino Leonardo to determine and adjust the system's power state.

## PUBLICATIONS

**Deep Learning Models Integrating Attention Mechanisms For Military Camouflaged Object Detection** — Nilgün Şengöz, Gül Karaman, Mert Samet Çeliker, Nazmi Yücel Çan — *El-Cezeri Fen ve Mühendislik Dergisi (2026)* [Link]

## EDUCATION

**Bachelor of Science in Information Systems and Technologies** — Burdur Mehmet Akif Ersoy University (GPA: 3.17/4.00)

09/2022 – 06/2026